

DATAWAREHOUSING AND ONLINE ANALYTICAL PROCESSING

SESSION 2
COMP6140001 – DATA MINING

SUBJECT MATTER EXPERT
D6667-Said Achmad



LEARNING OBJECTIVE

L01 : Explain concept of data, data warehouse, data preprocessing, and data mining



OUTLINE

- Data Warehouse: Basic Concepts
- OLTP vs OLAP
- Data Warehouse Modeling

DATAWAREHOUSE: BASIC CONCEPTS



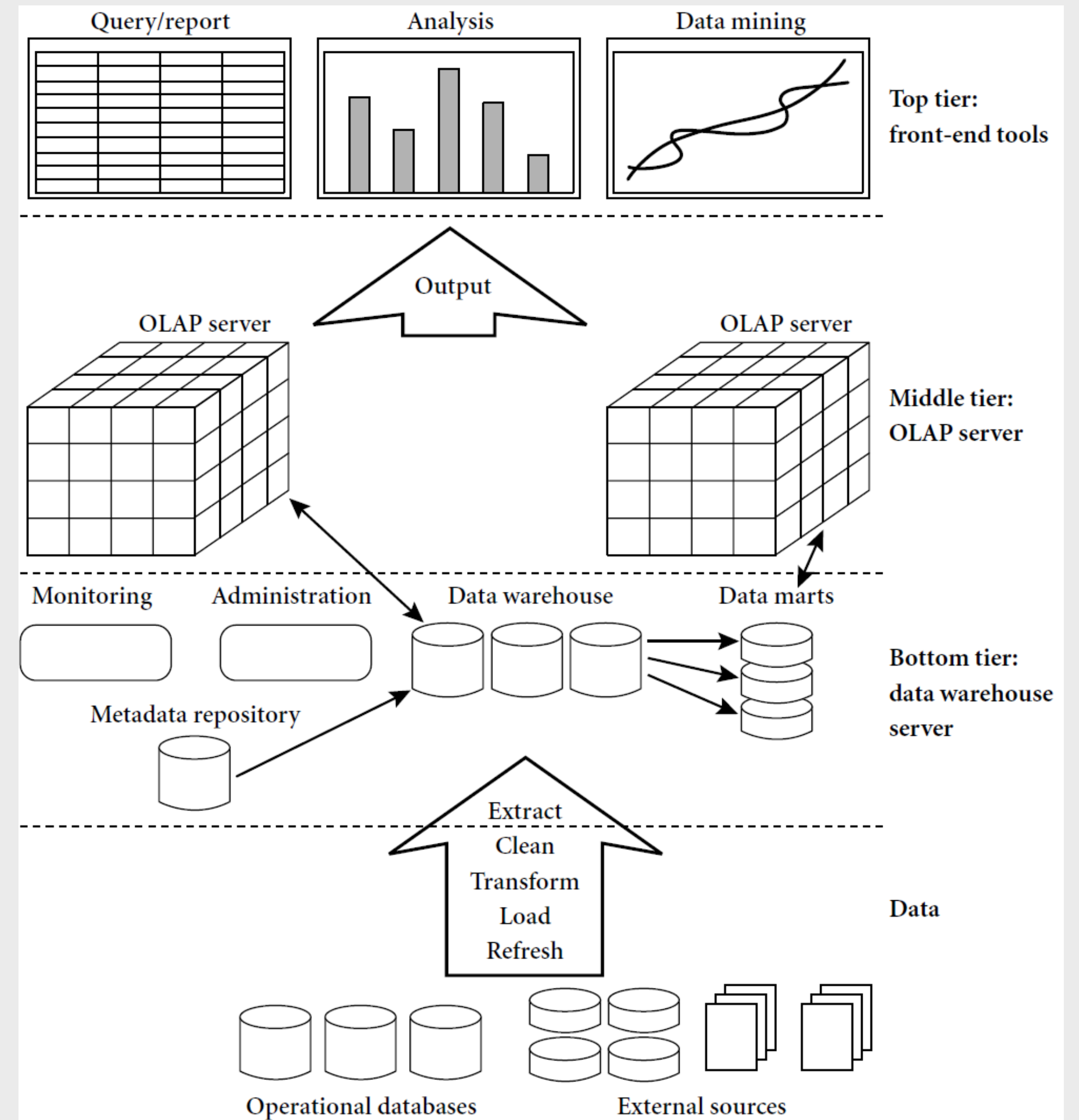
WHAT IS DATA WAREHOUSE?

- Defined in many different ways, but not rigorously.
 - A decision support database that is maintained **separately** from the organization's operational database
 - Support **information processing** by providing a solid platform of consolidated, historical data for analysis.
- “A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management's decision-making process.”—W. H. Inmon
- Data warehousing:
 - The process of constructing and using data warehouses



DATA WAREHOUSE: A MULTI-TIERED ARCHITECTURE

- Top Tier: Front-End Tools
- Middle Tier: OLAP Server
- Bottom Tier: Data Warehouse Server
- Data





DATA WAREHOUSE—SUBJECT-ORIENTED

- Organized around major subjects, such as **customer, product, sales**
- Focusing on the modeling and analysis of data for decision makers, not on daily operations or transaction processing
- Provide **a simple and concise** view around particular subject issues by **excluding data that are not useful in the decision support process**



DATA WAREHOUSE—INTEGRATED

- Constructed by integrating multiple, heterogeneous data sources
 - relational databases, flat files, on-line transaction records
- Data cleaning and data integration techniques are applied.
 - Ensure consistency in naming conventions, encoding structures, attribute measures, etc. among different data sources
 - E.g., Hotel price: currency, tax, breakfast covered, etc.
 - When data is moved to the warehouse, it is converted.



DATA WAREHOUSE—TIME VARIANT

- The time horizon for the data warehouse is significantly longer than that of operational systems
 - Operational database: current value data
 - Data warehouse data: provide information from a historical perspective (e.g., past 5-10 years)
- Every key structure in the data warehouse
 - Contains an element of time, explicitly or implicitly
 - But the key of operational data may or may not contain “time element”



DATA WAREHOUSE—NONVOLATILE

- A **physically separate store** of data transformed from the operational environment
- Operational **update of data does not occur** in the data warehouse environment
 - Does not require transaction processing, recovery, and concurrency control mechanisms
 - Requires only two operations in data accessing:
 - *initial loading of data* and *access of data*



DATA WAREHOUSE VS. HETEROGENEOUS DBMS

- Traditional heterogeneous DB integration: A **query driven** approach
 - Build **wrappers/mediators** on top of heterogeneous databases
 - When a query is posed to a client site, a meta-dictionary is used to translate the query into queries appropriate for individual heterogeneous sites involved, and the results are integrated into a global answer set
 - Complex information filtering, compete for resources
- Data warehouse: **update-driven**, high performance
 - Information from heterogeneous sources is integrated in advance and stored in warehouses for direct query and analysis

OLTP VS OLAP



DATA WAREHOUSE VS. OPERATIONAL DBMS

- OLTP (on-line transaction processing)
 - Major task of traditional relational DBMS
 - Day-to-day operations: purchasing, inventory, banking, manufacturing, payroll, registration, accounting, etc.
- OLAP (on-line analytical processing)
 - Major task of data warehouse system
 - Data analysis and decision making
- Distinct features (OLTP vs. OLAP):
 - User and system orientation: customer vs. market
 - Data contents: current, detailed vs. historical, consolidated
 - Database design: ER + application vs. star + subject
 - View: current, local vs. evolutionary, integrated
 - Access patterns: update vs. read-only but complex queries



OLTP VS. OLAP

	OLTP	OLAP
users	clerk, IT professional	knowledge worker
function	day to day operations	decision support
DB design	application-oriented	subject-oriented
data	current, up-to-date detailed, flat relational isolated	historical, summarized, multidimensional integrated, consolidated
usage	repetitive	ad-hoc
access	read/write index/hash on prim. key	lots of scans
unit of work	short, simple transaction	complex query
# records accessed	tens	millions
#users	thousands	hundreds
DB size	100MB-GB	100GB-TB
metric	transaction throughput	query throughput, response



WHY A SEPARATE DATA WAREHOUSE?

- High performance for both systems
 - DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
 - Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- Different functions and different data:
 - missing data: Decision support requires historical data which operational DBs do not typically maintain
 - data consolidation: DS requires consolidation (aggregation, summarization) of data from heterogeneous sources
 - data quality: different sources typically use inconsistent data representations, codes and formats which have to be reconciled
- Note: There are more and more systems which perform OLAP analysis directly on relational databases

DATA WAREHOUSE MODELING



FROM TABLES AND SPREADSHEETS TO DATA CUBES

- A data warehouse is based on a **multidimensional data model** which views data in the form of a data cube
- A data cube, such as **sales**, allows data to be modeled and viewed in multiple dimensions
- Dimension tables, such as **item (item_name, brand, type)**, or **time(day, week, month, quarter, year)**
- Fact table contains measures (such as **dollars_sold**) and keys to each of the related dimension tables
- In data warehousing literature, an n-D base cube is called a **base cuboid**. The top most 0-D cuboid, which holds the highest-level of summarization, is called the **apex cuboid**. The lattice of cuboids forms a **data cube**.

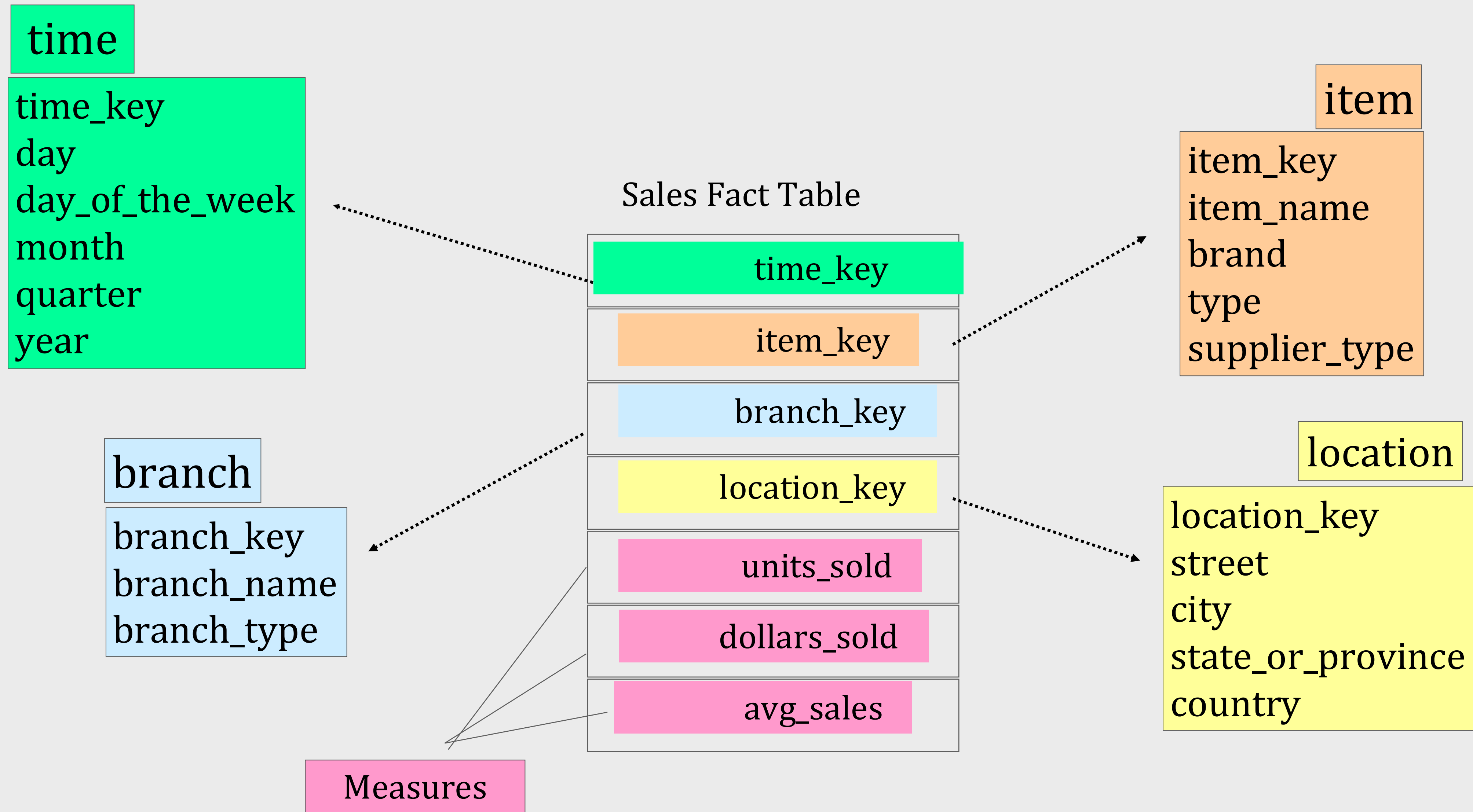


CONCEPTUAL MODELING OF DATA WAREHOUSES

- Modeling data warehouses:
 - Star schema: A fact table in the middle connected to a set of dimension tables
 - Snowflake schema: A refinement of star schema where some dimensional hierarchy is normalized into a set of smaller dimension tables, forming a shape similar to snowflake
 - Fact constellations: Multiple fact tables share dimension tables, viewed as a collection of stars, therefore called galaxy schema or fact constellation

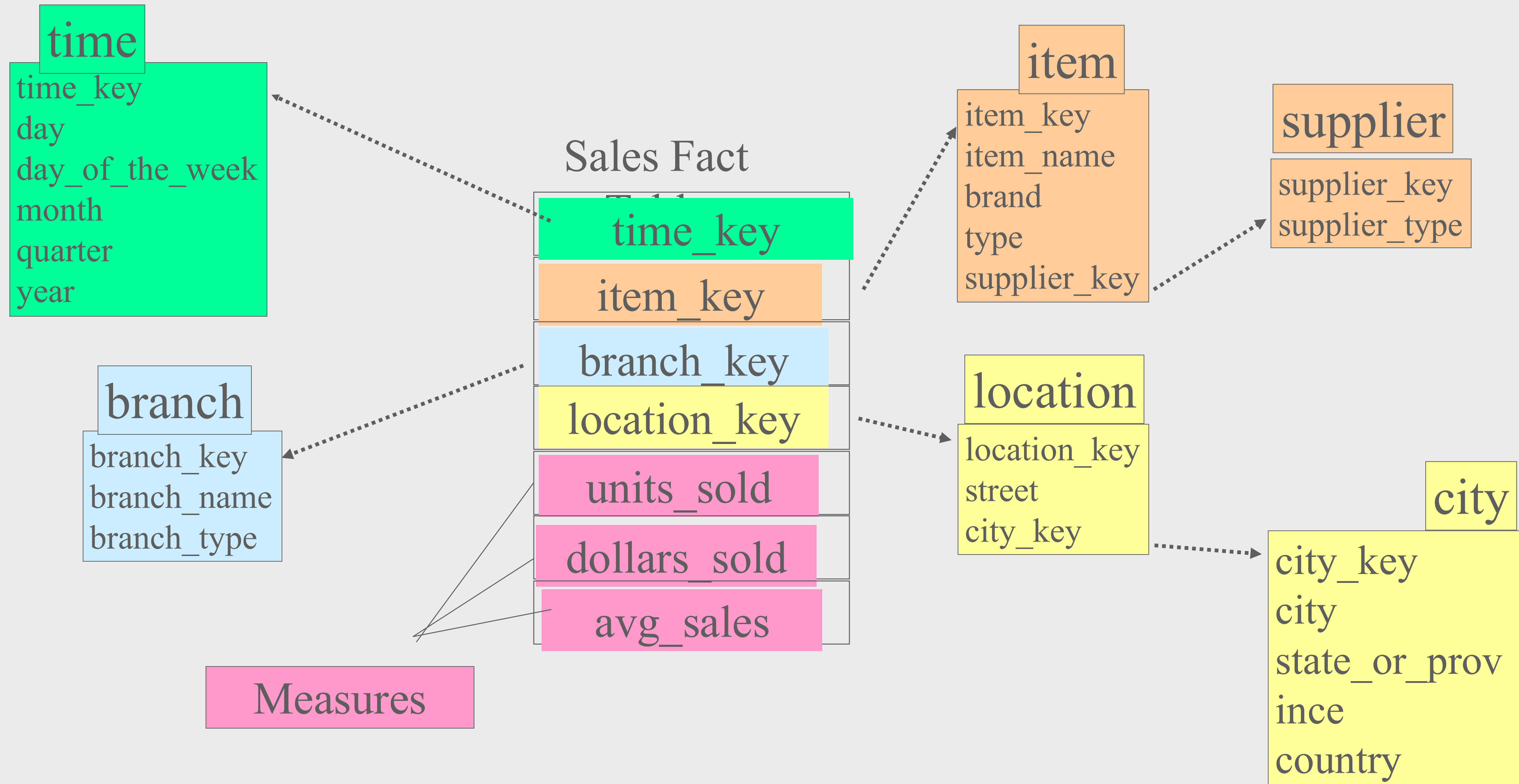


EXAMPLE OF STAR SCHEMA



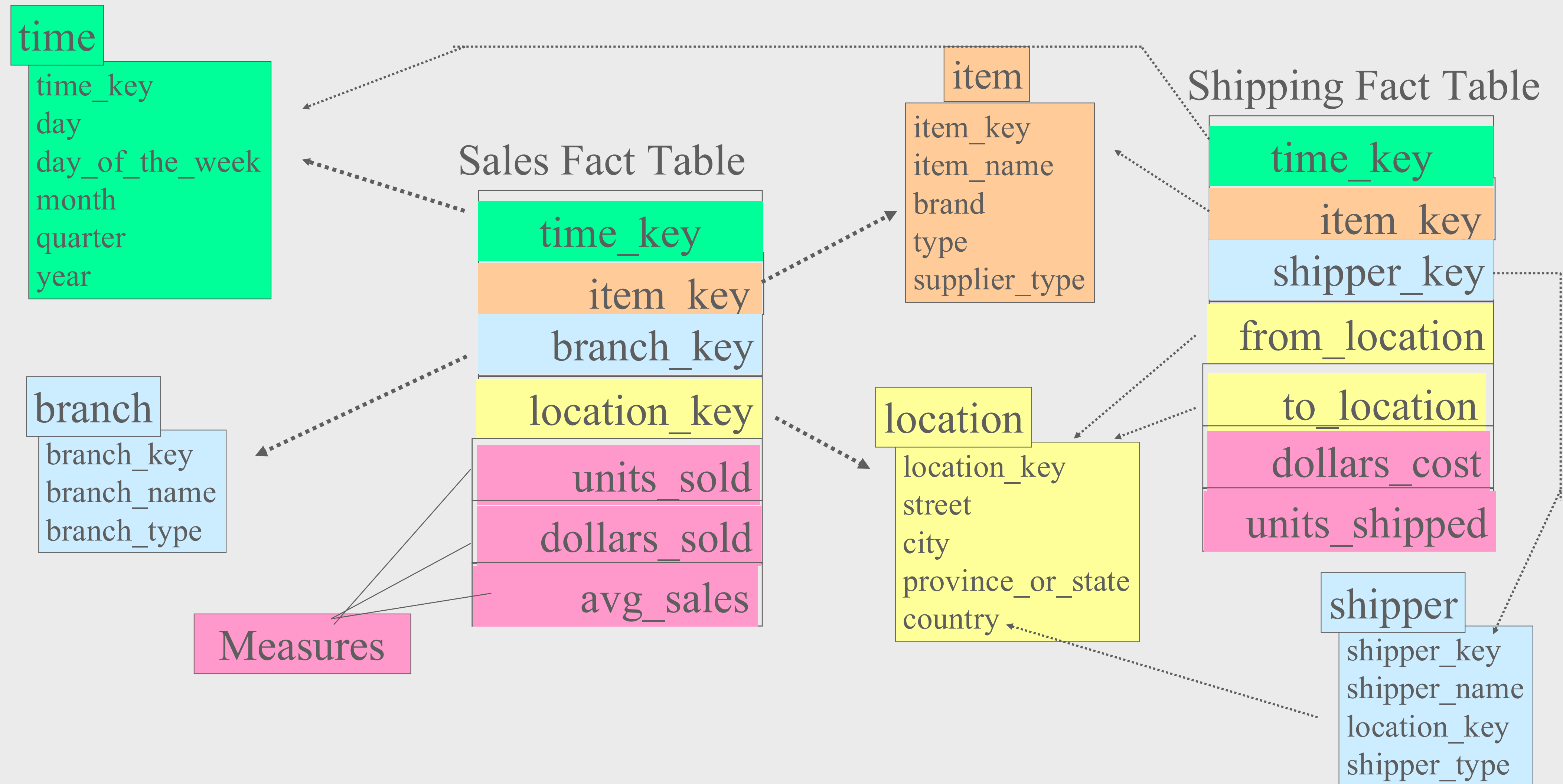


EXAMPLE OF SNOWFLAKE SCHEMA





EXAMPLE OF FACT CONSTELLATION





JOIN INDEX EXAMPLE

Dimension table: locations

Location_id	City	Province	...
L08	Vancouver	BC	...
L12	Victory	BC	...

Fact table: sales

TID	Location_id	Item_id	...	Amount
T57	L08	I8569	...	849.98
T238	L12	I7253	...	1099.95
T459	L12	I8569	...	799.95
T884	L08	I3051	...	2799.99

Dimension table: items

Item_id	Model	category	...
I7253	iPhone	smartphone	...
I8569	Galaxy	smartphone	...
I9051	iMac	Desktop	...

Join index table for
location-sales

Location_id	TID
L08	T57
L08	T884
L12	T238
L12	T459

Join index table for
item-sales

Location_id	TID
I7253	T238
I8569	T57
I8569	T459
I9051	T884

SUMMARY

- “A data warehouse is a subject-oriented, integrated, time-variant, and nonvolatile collection of data in support of management’s decision-making process.”
- DBMS— tuned for OLTP: access methods, indexing, concurrency control, recovery
- Warehouse—tuned for OLAP: complex OLAP queries, multidimensional view, consolidation
- Modeling data warehouses: Star schema, Snowflake schema, Fact constellations

EXERCISE

1. Please explain the difference between OLAP and OLTP
2. What is the relationship between Data Warehouse and OLAP?
3. Explain the difference between OLTP vs OLAP?

REFERENCES

Han, J., Kamber, M., & Pei, J. (2023). Data Mining: Concepts and Techniques (4rd ed.). San Francisco, CA, USA: Morgan Kaufmann Publishers Inc.